

**Amrita School of Engineering,
Bangalore-35
23MAT117-Linear Algebra
Lab Session Sheet -9**

Similar matrices, Diagonalization, Cayley Hamilton Theorem

- If $B=M^{-1}AM$, then A and B are similar. Similar matrices have same eigenvalues.

```
A = randi([0,9],5,5)
M = randi([0,9],5,5)
B = inv(M)*A*M
EvalA = eig(A)
EvalB = eig(B)
```

- AB and BA are similar and so have same non-zero eigenvalues

```
A=randi([0,9],5,3);
B=randi([0,9],3,5);
EvalAB=eig(A*B)
EvalBA=eig(B*A)
```

- $A^T A$ and AA^T are similar and have same non-zero eigenvalues.

```
A=randi([0,9],7,5);
EvalATA=eig(A'*A)
EvalAAT=eig(A*A')
```

- Application of diagonalization

```
Ld1=1;
Ld2=2;
Ld3=3;
Ev1 = [1; 1; 1];
Ev2 = [2; 1; -1];
Ev3 = [1; 2; -1];
V = [Ev1, Ev2, Ev3]
D = diag([Ld1, Ld2, Ld3])
A = V*D*inv(V)% Generates a matrix A with given eigenvalues Ld1, Ld2, Ld3.
EigValA = eig(A)
```

Cayley Hamilton theorem: Every square matrix satisfies its own characteristic equation.

It has two forms: Consider 3×3 matrix A . Let its characteristic equation be

$$|A - \lambda I| = \lambda^3 + a\lambda^2 + b\lambda + c = 0$$

1. $A^3 + aA^2 + bA + cI_3 = 0_{3 \times 3}$
2. $(A - \lambda_1 I)(A - \lambda_2 I)(A - \lambda_3 I) = 0_{3 \times 3}$

Here $\lambda_1, \lambda_2, \lambda_3$ are distinct (assumed) non-zero eigenvalues of matrix A

It can be verified that

$$\text{rank}(A - \lambda_1 I) = 2; \text{rank}(A - \lambda_2 I) = 2; \text{rank}(A - \lambda_3 I) = 2;$$

$$\text{rank}((A - \lambda_1 I)(A - \lambda_2 I)) = 1;$$

$$\text{rank}((A - \lambda_1 I)(A - \lambda_2 I)(A - \lambda_3 I)) = 0.$$

This result can be used to find 2 matrices whose product has a rank less than the rank of both the matrices.

Another observation:

Consider 3×3 lower triangular matrix with rank 2 and with all eigen values as zero.

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \text{ then}$$

$$A^2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^3 = A \times A^2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} \times \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Practice Questions:

- 1) Find a matrix A whose eigenvalues are 75 and 15 and eigenvectors are $[1,1]^T$ and $[1,-1]^T$. Also use diagonalization to find A^5 .
- 2) Generate a matrix with eigenvalues as 3,5,7,11 and eigenvectors as $[1,1,2,2]$, $[1,0,1,0]$, $[2,1,2,0]$ and $[0,1,1,0]$.
- 3) Find the eigenvalues of AA^T and $A^T A$ and verify both matrices are similar if $A = \begin{bmatrix} 1 & 3 \\ 0 & 4 \\ 1 & 6 \\ 2 & 7 \end{bmatrix}$.
- 4) Consider the matrix, $A = \begin{bmatrix} -4 & 1 & 1 \\ 1 & -6 & 7 \\ 1 & 7 & -9 \end{bmatrix}$. Find two matrices B and C that are similar to matrix A and verify their eigenvalues are same.
- 5) Create a 10×10 non-diagonal matrix with eigenvalues as 1, 2, ..., 10.
[Hint: Create a similar matrix using a non-singular matrix B :
 $A = B * \text{diag}([1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10]) * \text{inv}(B)$]
- 6) Create a 4×4 non-diagonal matrix with eigenvalues as 1,2,3,4. Find the following,
 - a) Rank of $A-I$, $A-2I$, $A-3I$, $A-4I$,
 - b) Rank of $(A-I)*(A-2I)$
 - c) Rank of $(A-I)*(A-2I)*(A-3I)$
 - d) Rank of $(A-I)*(A-2I)*(A-3I)*(A-4I)$
 and note down the observation.
- 7) Create 4×4 matrices with
 - (i) only one eigenvalue as 0,
 - (ii) only two eigenvalues as 0,
 - (iii) only three eigenvalues as 0.
 Find the rank in each case and comment on the change in rank.
- 8) Create $2 \ 9 \times 9$ matrices each with rank 7 using the concept of Cayley Hamilton Theorem mentioned above. What is the rank of the product of these two matrices.
- 9) Create a 4×4 matrix A , for which $A^4 = \mathbf{0}$ matrix.
- 10) Using MATLAB generate a 10×5 matrix A of rank 3.
 - (a) Obtain a symmetric matrix $S_1 = A * A^T$ and $S_2 = A^T * A$
 - (b) Find the rank of S_1 and S_2 .
 - (c) Find the eigenvalues of S_1 and S_2 . What do you notice about them? Explain.